

Effect of dynamical-photochemical coupling on oxygen airglow emission and implications for daytime ozone retrieved from 1.27 μm emission

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[1] A one-dimensional photochemical-diffusive-advective model is used to quantitatively investigate the effects of dynamical-photochemical coupling on the $\text{O}_2(\text{a}^1\Delta_g)$ concentration and the derived O_3 concentration based on the 1.27- μm airglow emission in the upper mesosphere and lower thermosphere. We compare the fractional differences of $\text{O}_2(\text{a}^1\Delta_g)$ and O_3 concentrations between those derived from the coupled model and from a photochemical steady state model that is often used for retrievals by airglow emissions. It is found that the fractional differences resulting from the use of a steady state model have a strong local time dependence when the chemical/radiative lifetime of $\text{O}_2(\text{a}^1\Delta_g)$ is comparable to that of diurnal variations and the transport timescale of tidal waves. In the upper mesosphere and lower thermosphere, especially near the mesopause region, the fractional difference may range from more than 100% in early morning immediately after sunrise to about 10% or less in the later afternoon.

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1. Introduction

[2] Airglow emissions are routinely used to remotely retrieve the abundances of atmospheric species. One of the brightest emission features in the mesosphere is the molecular oxygen dayglow at 1.27 μm resulting from the electronically excited molecular oxygen $\text{O}_2(\text{a}^1\Delta_g)$. $\text{O}_2(\text{a}^1\Delta_g)$ in the mesosphere is mainly produced by the photo-dissociation of ozone (O_3) in the Hartley band (200–310 nm), which makes the retrieval of O_3 from $\text{O}_2(\text{a}^1\Delta_g)$ photochemically feasible [e.g., Thomas *et al.*, 1984; Sica, 1991; Mlynchzak *et al.*, 1993; Yankovsky and Manuilova, 2006]. Previously, the retrieval algorithms assumed photochemical equilibrium under which mesospheric O_3 can be inferred by the volume emission rates at 1.27 μm [Thomas *et al.*, 1984]. In the lower thermosphere, collisional quenching of $\text{O}(\text{D})$ to $\text{O}_2(\text{b}^1\Sigma_g)$ and then from $\text{O}_2(\text{b}^1\Sigma_g)$ to $\text{O}_2(\text{a}^1\Delta_g)$ is the main source for $\text{O}_2(\text{a}^1\Delta_g)$ production [Mlynchzak *et al.*, 1993].

[3] Because the strength of airglow emissions by $\text{O}_2(\text{a}^1\Delta_g)$ is proportional to the populations of the excited states of the molecular oxygen, the accuracy of the airglow modeling that involves the species of interest (O_3 in this case) depends critically on the photochemical timescales of those excited states [e.g., Zhu and Yee, 2007]. Furthermore, when the photochemical timescales of the species in their excited states are comparable to the transport timescales, photochemical and transport processes can be strongly

coupled [e.g., Zhu *et al.*, 2000; Brasseur and Solomon, 2005]. An airglow model not accounting for time-dependent photochemical coupling among different chemical species and not including the dynamical coupling between transport and photochemical processes could lead to nonnegligible errors in species retrieval.

[4] It is known that rapidly varying photochemical processes are indirectly coupled with transport by slowly varying photochemistry. Several coupling techniques used in models are discussed by Zhu *et al.* [2000], including the most familiar approach of family species as a special case. The rapidly varying species within a family can be assumed to be in photochemical equilibrium and be calculated independently from the transport processes. Similarly, given the modeled or measured chemical compositions of various species, the vibrationally or electronically excited states can be approximated off-line by assuming local chemical equilibrium. As a consequence, the relevant chemical species can be inferred photochemically with an equilibrium model from volume emission rates that are directly related to the number densities of the excited states. On the other hand, for airglow originating from metastable electronic transitions, such as $\text{O}_2(\text{a}^1\Delta_g \rightarrow \text{X}^1\Sigma_g)$, which produces the 1.27- μm airglow emission, the spontaneous relaxation times at high altitudes are usually longer than the photochemical times of the relevant chemical species and could be comparable to the diurnal variation and transport timescale, leading to a strong coupling between the photochemistry forced by the diurnally varying photolysis rates, the advective transport by the tidal velocities, and the diurnal variation in rate coefficients induced by the tidal temperature perturbations in the mesosphere and lower thermosphere [Zhu *et al.*, 1999a, 2000].

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[5] Several studies have examined the sensitivity of the O₃ concentration with respect to some of the model parameters and model assumptions. Using a steady state photochemical model, *Mlynczak et al.* [1993], *Mlynczak and Olander* [1995], and *Mlynczak and Nesbitt* [1995] investigated the effects of kinetic and spectroscopic rate coefficients on the accuracy of the inferred O₃ concentration. Recently, *Yankovsky and Manuilova* [2006] found noticeable differences in the volume emission rates at 762 nm and 1.27 μm while explicitly including the electronic-vibrational kinetics of the excited products in a more comprehensive photochemical steady state model for the O₂ airglow emissions. A time-dependent photochemical model was used by *Sica* [1991] to study the sensitivity of the retrieved O₃ profiles to uncertainties in various model parameters such as kinetic rate coefficients, temperature profiles, solar flux, and the time-dependent nature of the photochemistry. Because the model was developed for inferring mesospheric ozone from ground-based measurements of oxygen dayglow near twilight, the effect of the long radiative relaxation time of O₂(a¹ Δ_g) on the O₃ retrieval was examined by comparing the difference in O₂(a¹ Δ_g) profiles between sunrise and sunset [*Sica*, 1991].

[6] This paper systematically addresses potential errors in O₃ retrieval based on the airglow emissions of the O₂(¹ Δ) band at 1.27 μm and will also briefly describe errors in O₃ retrieval based on the O₂(¹ Σ) band at 762 nm. Specifically, we will introduce two sensitivity coefficients that relate fractional errors in O₃ to fractional errors in O₂(¹ Δ) and O₂(¹ Σ). This will allow us to separate and quantify different sources of errors that determine the chemical and radiative feasibility of deriving one species from another. Furthermore, we will only focus on the O₃ retrievals in the mesosphere and lower thermosphere. Section 2 briefly describes the one-dimensional photochemical-diffusive-advective model used for the current study. We use sensitivity coefficients that relate the fractional errors between two quantities to illustrate the error amplification in retrievals. Section 3 presents the numerical results that show issues involved in deriving O₃ from the measured O₂(¹ Δ) emission at 1.27 μm near sunrise, especially due to the large uncertainty of the tidal waves in the upper mesosphere. Conclusions are given in section 4.

2. One-Dimensional Airglow Model and Sensitivity Coefficients

[7] We adopt an updated version of the JHU/APL one-dimensional (1D) photochemical-diffusive-advective model previously developed for modeling the diurnal variation of ozone photochemistry in the upper middle atmosphere [*Zhu et al.*, 2000, 2003]. The input fields of the model are the slowly varying climatology of temperature, meridional circulation, and tracer distributions derived from a coupled two-dimensional model [*Zhu et al.*, 1997, 1999a] and the tidal fields derived from a spectral tidal model [*Zhu et al.*, 1999b]. To include the molecular oxygen airglow emissions, we added additional kinetic and radiative processes that produce and destroy O₂(¹ Δ) and O₂(¹ Σ) to the photochemical solver. The added photochemical processes, shown in Table 1, mostly follow the schemes used by *Mlynczak et al.* [1993]. In our model, we have included

the photodissociation of O₃ into O(¹D) and O₂, i.e., reaction J_{z0} in Table 1 [*Brasseur and Solomon*, 2005], which was not included in several previous photochemical models of the O₂ airglow emissions [e.g., *Thomas et al.*, 1984; *Sica*, 1991; *Mlynczak et al.*, 1993]. Because the time-dependent integration of the model will encounter a long absorber slant path near twilight we have also added near-infrared absorption of O₂ at 1.27 μm (J_{z4}), with the absorption cross section taken from the recent measurements by *Smith and Newnham* [2000]. For the convenience of presenting the analytic solutions of the photochemical equilibrium model and sensitivity studies, we have grouped some photochemical processes used by *Mlynczak et al.* [1993] into composite reactions with effective rate coefficients and fictitious molecules. For example, the loss of O₂(¹ Σ) or the production of O₂(¹ Δ) through collisional quenching of O₂(¹ Σ) to O₂(¹ Δ) as symbolically expressed by the reaction k_{z4} can be written as

$$k_{z4}M_3[O_2(^1\Sigma)] \equiv [O_2(^1\Sigma)]\{k_{12}[N_2] + k_{13}[CO_2] + k_{14}[O_3] + k_{15}[O] + k_{16}[O_2]\}, \quad (1)$$

where the rate coefficients k_{12} through k_{16} are for reactions 12 through 16 listed in Table 1 of *Mlynczak et al.* [1993]. Similarly, we also denote

$$k_{z2}M_1[O(^1D)] \equiv [O(^1D)]\{k_5[N_2] + k_7[O_2]\}, \quad (2)$$

$$k_{z3}M_2[O_2(^1\Delta)] \equiv [O_2(^1\Delta)]\{k_8[O_2] + k_9[N_2] + k_{10}[O]\}. \quad (3)$$

By symbolically grouping similar reactions, we have reduced the 12 photochemical reactions used by *Mlynczak et al.* [1993] to 4 reactions, as shown in Table 1 for J_{z2} and for k_{z2} through k_{z4} . The assumption of photochemical equilibrium for O(¹D), O₂(¹ Σ), and O₂(¹ Δ) leads to the following set of equations for three unknown densities:

$$O(^1D) : [O(^1D)](k_{z1}[O_2] + k_{z2}M_1) = (J_{z0} + J_{z1})[O_3] + J_{z2}[O_2], \quad (4)$$

$$O_2(^1\Sigma) : [O_2(^1\Sigma)](k_{z4}M_3 + A_{z2}) = [O_2](J_{z3} + k_{z1}[O(^1D)]), \quad (5)$$

$$O_2(^1\Delta) : [O_2(^1\Delta)](k_{z3}M_2 + A_{z1}) = J_{z1}[O_3] + J_{z4}[O_2] + k_{z4}M_3[O_2(^1\Sigma)]. \quad (6)$$

The number densities [O(¹D)], [O₂(¹ Σ)], and [O₂(¹ Δ)] can be solved from equations (4)–(6) consecutively and expressed as functions of the basic state, such as [N₂], [O₂], and other species such as [O₃] and [O]. From equation (4) it immediately becomes clear that the inclusion of J_{z0} will lead to an increase in O(¹D) by $\sim 10\%$ below about 90 km, where O₃ photolysis is the major source of O(¹D) [*Mlynczak et al.*, 1993].

[8] A significant portion of available mesospheric daytime O₃ measurements have been obtained by using the airglow emissions at 1.27 μm from O₂(¹ Δ) [e.g., *Thomas*,

Table 1. Photochemical Reactions in the Model

Rate Symbol	Process	Rate or Efficiency
J_{z0}	$O_3 - h\nu \rightarrow O(^1D) + O_2$	$1 - \phi = 0.1$
J_{z1}	$O_3 - h\nu \rightarrow O(^1D) + O_2(^1\Delta)$	$\phi = 0.9$
J_{z2}	$O_2 - h\nu \rightarrow O(^1D) + O(^3P)$	Ly α and SRC
J_{z3}	$O_2 - h\nu(762 \text{ nm}) \rightarrow O_2(^1\Sigma)$	$k(\infty) = 5.35 \times 10^{-9} \text{ s}^{-1}$
J_{z4}	$O_2 - h\nu(1.27 \text{ }\mu\text{m}) \rightarrow O_2(^1\Delta)$	$k(\infty) = 1.54 \times 10^{-10} \text{ s}^{-1}$
A_{z1}	$O_2(^1\Delta) \rightarrow O_2 + h\nu(1.27 \text{ }\mu\text{m})$	$k = 2.237 \times 10^{-4} \text{ s}^{-1}$
A_{z2}	$O_2(^1\Sigma) \rightarrow O_2 + h\nu(762 \text{ nm})$	$k = 0.0758 \text{ s}^{-1}$
k_{z1}	$O(^1D) + O_2 \rightarrow O(^3P) + O_2(^1\Sigma)$	$k = 2.5 \times 10^{-11} \times \exp(70/T) \text{ cm}^3 \text{ s}^{-1}$
k_{z2}	$O(^1\Delta) + M_1 \rightarrow O(^3P) + M_1$	composite k [Mlyneczek <i>et al.</i> , 1993]
k_{z3}	$O_2(^1\Delta) + M_2 \rightarrow O_2 + M_2$	composite k [Mlyneczek <i>et al.</i> , 1993]
k_{z4}	$O_2(^1\Sigma) + M_3 \rightarrow O_2(^1\Delta) + M_3$	composite k [Mlyneczek <i>et al.</i> , 1993]

1990]. O₃ can also be retrieved by the 762-nm emission from O₂(¹Σ) [Yee *et al.*, 1993; Mlyneczek *et al.*, 2001; Marsh *et al.*, 2002]. To quantitatively describe the advantages and disadvantages of these two retrieval techniques it is worthwhile investigating the parameter sensitivities between O₂(¹Δ), O₂(¹Σ), and O₃. We define the sensitivity parameters S_{13} and S_{23} to be the coefficients that relate the fractional variations in [O₃] with respect to that in [O₂(¹Δ)] and [O₂(¹Σ)], respectively:

$$\frac{\delta[O_3]}{[O_3]} = S_{13} \frac{\delta[O_2(^1\Delta)]}{[O_2(^1\Delta)]}, \quad (7)$$

$$\frac{\delta[O_3]}{[O_3]} = S_{23} \frac{\delta[O_2(^1\Sigma)]}{[O_2(^1\Sigma)]}. \quad (8)$$

The ideal value of S_{13} or S_{23} for an accurate retrieval of O₃ from airglow emission either by O₂(¹Δ) or O₂(¹Σ) is 1, which represents a perfect correlation. A good example of such a perfect correlation is the direct proportionality of the O₂ volume emission rate V_{z1} at 1.27 μm to [O₂(¹Δ)] under the optically thin limit:

$$V_{z1} = A_{z1} [O_2(^1\Delta)], \quad (9)$$

where A_{z1} is the Einstein coefficient for the spontaneous emission shown in Table 1. Assuming that A_{z1} is known accurately, we have

$$\frac{\delta[O_2(^1\Delta)]}{[O_2(^1\Delta)]} = \frac{\delta V_{z1}}{V_{z1}}, \quad (10)$$

i.e., it has a unit value in its sensitivity coefficient. A unit value of the sensitivity coefficient between [O₂(¹Δ)] and V_{z1} means a perfect correlation. A value much smaller than 1 for a sensitivity coefficient would imply an insensitivity of O₂(¹Δ) variability to the measured or the retrieved variation of the volume emission rate V_{z1} . On the other hand, a value much bigger than 1 in sensitivity coefficient would lead to a strong amplification of errors in the retrieved O₂(¹Δ) from errors in V_{z1} resulting from the measurement and inversion.

[9] Differentiating equations (4)–(6) with respect to three unknowns plus [O₃] and solving the resulting differentiating equations, we obtain

$$S_{13} = \left[J_{z1} + \frac{(J_{z0} + J_{z1})k_{z1}[O_2] - k_{14}K_1[O_2(^1\Sigma)]}{K_1[1 + R_{z2}/(k_{z4}M_3)]} \right]^{-1} \cdot (k_{z3}M_2 + A_{z1}) \frac{[O_2(^1\Delta)]}{[O_3]}, \quad (11)$$

$$S_{23} = \frac{K_1(k_{z4}M_3 + A_{z2})[O_2(^1\Sigma)]}{[O_3][(J_{z0} + J_{z1})k_{z1}[O_2] - k_{14}K_1[O_2(^1\Sigma)]]}. \quad (12)$$

where $K_1 \equiv k_{z1}[O_2] + k_{z2}M_1$ and k_{14} is defined in equation (1). S_{13} and S_{23} can also be derived numerically from their definitions in equations (7) and (8) by fractionally perturbing [O₃] in the original steady state model of equations (4)–(6). Unlike in equation (10) that has a constant coefficient, the sensitivity coefficients S_{13} and S_{23} are no longer constants but functions of the photochemical states of the system, which could have large spatial and temporal variability. Note that a value either much smaller (insensitive) or much larger (error amplifying) than 1 in S_{13} or S_{23} will lead to an ill-conditioned problem and thus lead to significant errors in O₃ retrievals.

[10] Sensitivity coefficients similar to those defined in equations (7) and (8) have been formally defined and used in other studies of O₃ measurements and modeling analyses [e.g., Keating *et al.*, 1987; Zhu *et al.*, 2003]. However, they have usually been defined in the context of the forcing-response relationship that describes changes of a photochemical system to a prescribed variation of an external forcing such as the solar flux. The sensitivity coefficient defined in this paper establishes a relationship of the fractional variations between two closely related quantities, and its usage is mainly associated with error analysis in retrieval problems. Note that qualitatively, [O₂(¹Δ)] in the mesosphere is sensitive to [O₃], which makes the retrieval of O₃ distribution from O₂(¹Δ) feasible. Our definition of S_{13} shown in equation (11) provides a quantitative measure of the feasibility of such an approach. Specifically, the special case of a unit value of the sensitivity coefficient between O₂(¹Δ) and V_{z1} as shown by equations (9) and (10) gives us a physical insight into its usage in two aspects: (1) variations in V_{z1} give the maximum sensitivity in

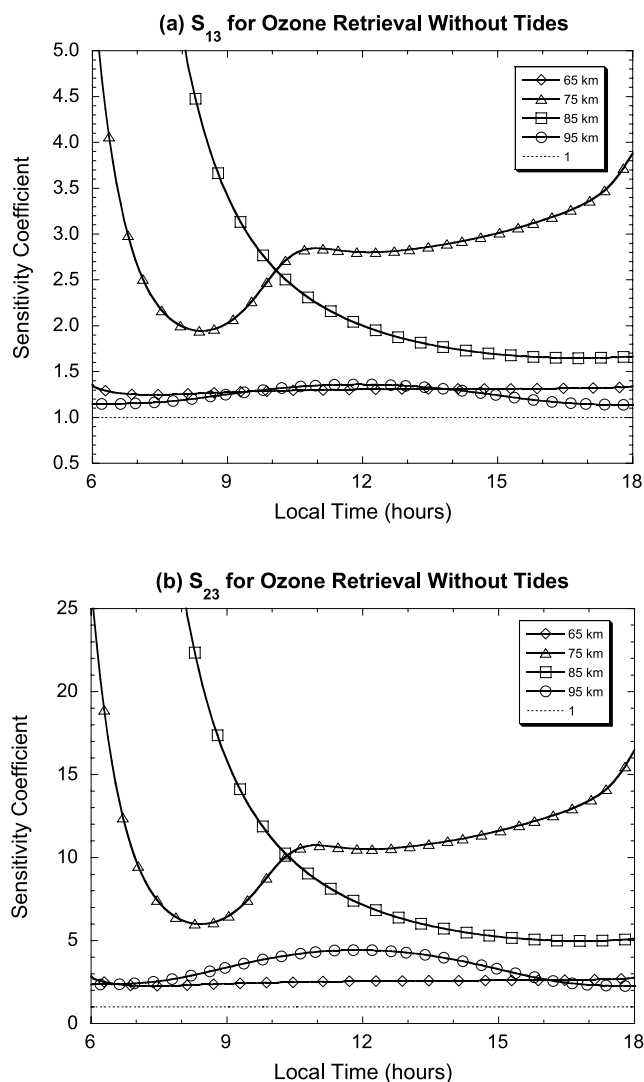


Figure 1. Sensitivity coefficients S_{13} and S_{23} for ozone retrieval at selected altitudes at the equator on 15 June as a function of local time. S_{13} and S_{23} are derived from the JHU/APL 1D photochemical-diffusive-advective model without tidal waves.

[O₂(¹Δ)] without an amplification of errors in retrieval and (2) errors in [O₂(¹Δ)] are solely dependent on errors in V_{z1} as long as equation (9) holds. These two noted points derived from the special case of equation (10) establishes a benchmark standard that can be used to help us to appreciate the physical significance of S_{13} and S_{23} .

[11] Figure 1 shows S_{13} and S_{23} at selected altitudes and at the equator as a function of local time. The JHU/APL 1D photochemical model was run without tides for 15 June, which is near the Northern Hemisphere summer solstice. We first note that the two panels in Figure 1 have different ordinates so that overall the sensitivity coefficient S_{23} is about a factor 5 greater than S_{13} in the upper mesosphere. This means that errors in the retrieved O₃ from the O₂(¹Σ) emission at 762 nm will be much more greatly amplified than O₃ from O₂(¹Δ) at 1.27 μm, assuming that the photochemical equilibrium model described by equations (4)–(6)

is a good approximation. This is consistent with the previous results derived from the error analysis when O₃ was derived from simultaneous measurements of O₂(¹Δ) and O₂(¹Σ) [Mlynchak *et al.*, 2001]. Physically, this also means that O₂(¹Δ) is largely determined by the direct O₃ photolysis (J_{z1} in Table 1) and thus is linearly proportional to [O₃] whereas O₂(¹Σ) is only indirectly related to [O₃], which leads to a much smaller S_{23}^{-1} in equation (8) than S_{13}^{-1} in equation (7). Since the sensitivity coefficients S_{13} and S_{23} as defined by equations (11) and (12) depend on the concentrations of minor species, they also show strong local time dependencies similar to the minor species. Note that a magnitude of 10 for S_{23} near the mesopause means that a 5% error in the inverted [O₂(¹Σ)] will be amplified to 50% in the O₃ retrieval. This is consistent with the similar finding by Mlynchak and Olander [1995] that the errors in kinetic and spectroscopic rates are significantly enhanced in the retrieved O₃ around 75 km. Our definitions of the sensitivity coefficients only provide the diagnostic relationships between two errors on the left and right sides of the expressions. The fractional errors on the right-hand sides of equations (7) and (8) could be due to the measurement and inversion that derive [O₂(¹Δ)] or [O₂(¹Σ)], due to the uncertainties within the steady state photochemical model (i.e., rate coefficients, etc), or due to the approximation of the modeling (i.e., neglect of transport, etc.). When the photochemical processes are dominant or comparable to the rest processes S_{13} and S_{23} are given by equations (11) and (12).

[12] Because S_{13} and S_{23} depend on the concentrations of various chemical species, these two sensitivity coefficients could also be dependent on model or parameterization when the chemical species are derived under different modeling settings. We have derived S_{13} and S_{23} in Figure 1 without including the effects of tidal wave motion, which periodically advects the species, and tidal temperature perturbation, which changes the kinetic rate coefficients. In the upper mesosphere near the equator, the diurnal tide is the dominant component in temperature and vertical velocity perturbations whereas the semidiurnal tide is the dominant component in horizontal wind perturbations. The tidal amplitudes for temperature and vertical velocity near 85 km are about 10 K and 10 cm s⁻¹, reaching their maximum values near dawn and noon, respectively [Zhu and Yee, 1999; Zhu *et al.*, 1999b]. In Figure 2, we show the same plots of S_{13} and S_{23} as in Figure 1 except that these tidal effects are included [Zhu *et al.*, 2000]. Comparison between Figure 1 and Figure 2 shows that the overall structure of the sensitivity coefficients remains the same except around 85 km when [O₃] is extremely low and the vertical gradient of [O₃] is large: (1) S_{23} is still about a factor of 5 greater than S_{13} , again indicating the same difficulty of O₃ retrieval by O₂(¹Σ) emission as described above; (2) the local time dependence of S_{13} and S_{23} remains approximately the same except near the mesopause and twilight when the advective transport greatly alleviates the rapid changes in concentrations and thus also the variations in the sensitivity coefficients; and (3) the maximum values of S_{13} and S_{23} occur near the mesopause around 80 km where O₃ has its minimum concentration. More detailed spatial and temporal distribution of S_{13} together with its quasi-invariance with respect to transport by tidal waves will be shown later.

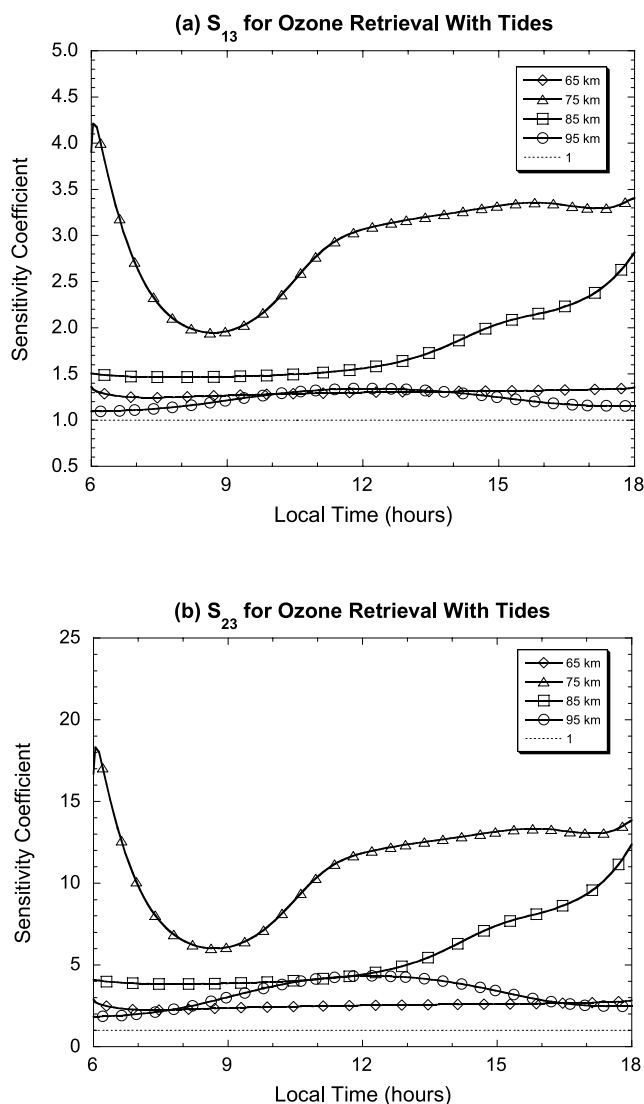


Figure 2. Same as in Figure 1 except the effects of tidal waves have been included in the model.

[13] The quasi-invariance of the sensitivity coefficients described above makes it possible to separate different sources of errors in O₃ retrievals using the airglow emissions when the errors are not too large in comparison with the retrieved O₃. Because of the very large value of S_{23} in the upper mesosphere, which makes the accurate O₃ retrieval by O₂(¹Σ) emission difficult, the following discussions will mainly focus on the O₃ retrieval using O₂(¹Δ) emissions at 1.27 μm. Given a volume emission rate V_{z1} retrieved from the observed slant airglow emission, one may immediately derive [O₂(¹Δ)] from equation (9) without error amplification. To invert [O₃] from [O₂(¹Δ)] we need an airglow model to simulate [O₂(¹Δ)] and to establish a relationship between [O₃] and [O₂(¹Δ)]. Currently, one of the comprehensive photochemical models for [O₂(¹Δ)] in the mesosphere and lower thermosphere is the one by Mlynarczyk *et al.* [1993], which contains recent updates to the kinetics and is expressed in equations (4)–(6) in this paper. Mlynarczyk and Olander [1995] and Mlynarczyk and Nesbitt [1995] used this model to conduct the error analysis

of O₃ retrieval to the uncertainties in different kinetic and spectral rate coefficients.

[14] The airglow model of equations (4)–(6) is a steady state photochemical model with only three unknowns. A more accurate model should include the time dependence of all the chemical species that vary on timescales similar to the three unknowns. Specifically, it should also include the diurnal variation because all three unknown species in equations (4)–(6) are forced by diurnally varying photolysis rates. It should also include both advective and diffusive transport if the transport timescales are also comparable to that of the diurnal variations. Mathematically, the complete continuity equation for a chemical species can be expressed as [e.g., Zhu *et al.*, 2000; Strobel, 2002; Zhu and Yee, 2007]

$$\frac{\partial \chi_i}{\partial t} = (P_i - L_i \chi_i) + T_i(\nabla \chi_i), \quad (13)$$

where t is time, P_i and $L_i \chi_i$ are the photochemical production and loss terms for species i of mixing ratio χ_i , respectively. $T_i(\nabla \chi_i)$ denotes the transport terms that include both advective and diffusive transport and are generally dependent on the gradient of the species mixing ratio $\nabla \chi_i$ [Zhu *et al.* 2000]. When P_i and $L_i \chi_i$ become dominant over the rest of the terms in equation (13) and form an approximate balance between the two we can use a photochemical steady state model such as equations (4)–(6) to solve for χ_i . When P_i and $L_i \chi_i$ are much smaller than the rest of the terms, which could occur when χ_i is small and $\nabla \chi_i$ is large, both the retrieved χ_i and the error analysis based on the photochemical equilibrium model become invalid or questionable. For a given species, such as O₂(¹Δ), its concentration can be derived by a photochemical steady state model, such as equations (4)–(6), when concentrations of O₃ and other necessary species are known. O₂(¹Δ) can also be derived by integrating the comprehensive continuity equation (13) along with all the other species. The difference between these two approaches gives a measure of how good the steady state model is for deriving O₂(¹Δ) when O₃ is given. The sensitivity coefficient S_{13} introduced in equation (7) yields the difference in O₃ between the two approaches when O₂(¹Δ) is given, as in the O₃ retrieval problem.

[15] In the current modeling problem of the time-dependent integration over the diurnal variation, the baseline or the basic timescale corresponding to the term $\partial \chi_i / \partial t$ is determined by the external forcing of the varying photolysis of a few minutes to a few hours. In the upper middle atmosphere where the advective transport by tidal waves becomes important [Zhu *et al.*, 2000] the transport timescale by tidal waves is a few hours. The photochemical timescales of various species are determined by collisional quenching and radiative relaxation. The relaxation time of the spontaneous emission of O₂(¹Δ) at 1.27 μm is 1.24 hours [Lafferty *et al.*, 1998; Spalek *et al.*, 1999]. Therefore, at high altitude where the collisional quenching becomes less important, the photochemical timescale of O₂(¹Δ) could be comparable or longer than the basic timescale and the transport timescale.

[16] Similar to the study of the O₃ sensitivity to different kinetic and spectral rate coefficients of a steady state model, we can also examine the model sensitivity in [O₃] and [O₂(¹Δ)] due to the inclusion of different physical

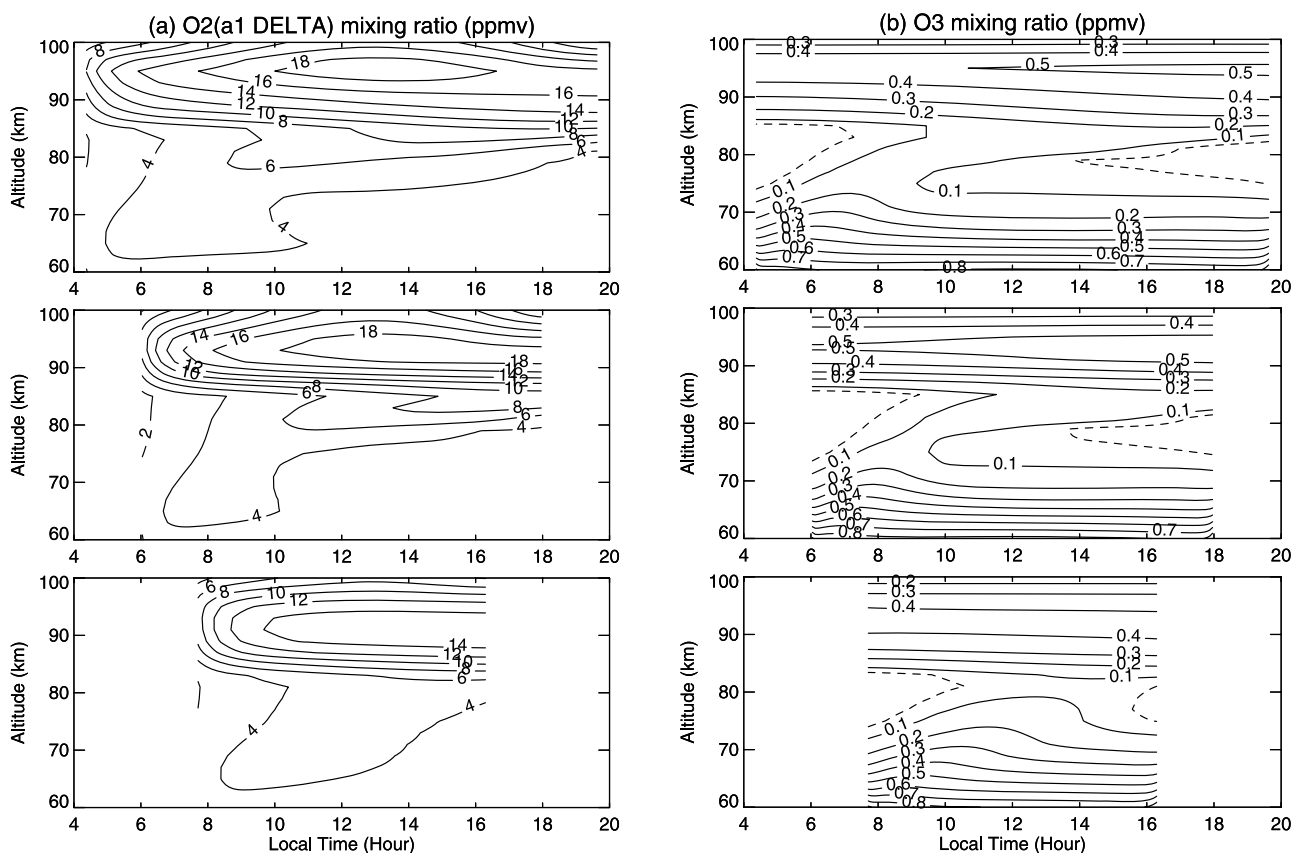


Figure 3. Modeled daytime (a) O₂(¹Δ) and (b) O₃ mixing ratios without tidal effects on 15 June as a function of altitude and local time at three latitudes: (top) 40°N, (middle) equator, and (bottom) 40°S.

processes, i.e., different terms in the complete continuity equation (13). In general, a model that includes more physical processes is expected to lead to a more accurate result if the added physical processes play important roles in modifying the tracer distributions and are of small uncertainties. In the current problem, the tidal waves have significant effects on tracer distributions in the upper mesosphere [e.g., Zhu *et al.*, 2000]. However, there are also great uncertainties in both tidal phase and amplitude in the upper mesosphere due to the uncertainties in tidal forcing and dissipation [e.g., Hagan, 2000; Talaat *et al.*, 2001]. The incompleteness and the uncertainties in modeling can all be treated the same as the uncertainties in kinetic and spectral rate coefficients in the photochemical model or the measurement and inversion errors in O₂(¹Δ) that ultimately lead to the errors in O₃ retrieval.

[17] Errors in retrieved O₃ due to different model specifications can be examined directly by comparing different model outputs of O₃, i.e., by taking the difference between O₃ derived from the steady state model (4)–(6) and the one from our fully coupled model (13). They can also be examined by comparing the different model outputs of O₂(¹Δ) and then converting errors in O₂(¹Δ) into errors in O₃ through the sensitivity relationship equation (7). The major advantages of decomposing the total error into two different sources are (1) more physical insights can be gained so that one may have a better idea of how to deal with or control the errors caused by different sources and (2) the error analysis or the sensitivity study becomes

simpler because the current decomposition of two steps is in accordance with the retrieval process of deriving [O₂(¹Δ)] from the volume emission rate V_{z1} followed by deriving [O₃] from O₂(¹Δ). In fact, our conclusion about deriving [O₃] from O₂(¹Σ) emission at 762 nm being a difficult problem near the mesopause, which was based on the results shown in Figures 1 and 2, has been derived from a single source of errors due to the introduction and analysis of S_{23} . Note that equation (7) establishes the relationship of the fractional variations between O₃ and O₂(¹Δ), which could correspond to fractional errors associated with uncertainties in measurements or modeling. In this paper, we examine the fractional differences of O₃ and O₂(¹Δ) resulting from different parameter settings of the photochemical model, such as with or without tidal forcing and/or a steady state assumption of the model. These fractional differences can be considered as fractional errors due to modeling if we assume that the species derived from a comprehensive model are the accurate solutions and the differences cannot be systematically corrected because of, say, the uncertainties in tidal forcing. For a systematic difference resulting from some parameter settings in a model, it can be corrected/eliminated through a modeling effort to improve the retrieved species.

[18] We have assumed implicitly an invariance in sensitivity coefficients when we argued the advantages of the decomposition in errors. In other words, a change in a model parameter that leads to a fractional difference in O₂(¹Δ) will not lead to a significant change in S_{13} . Our

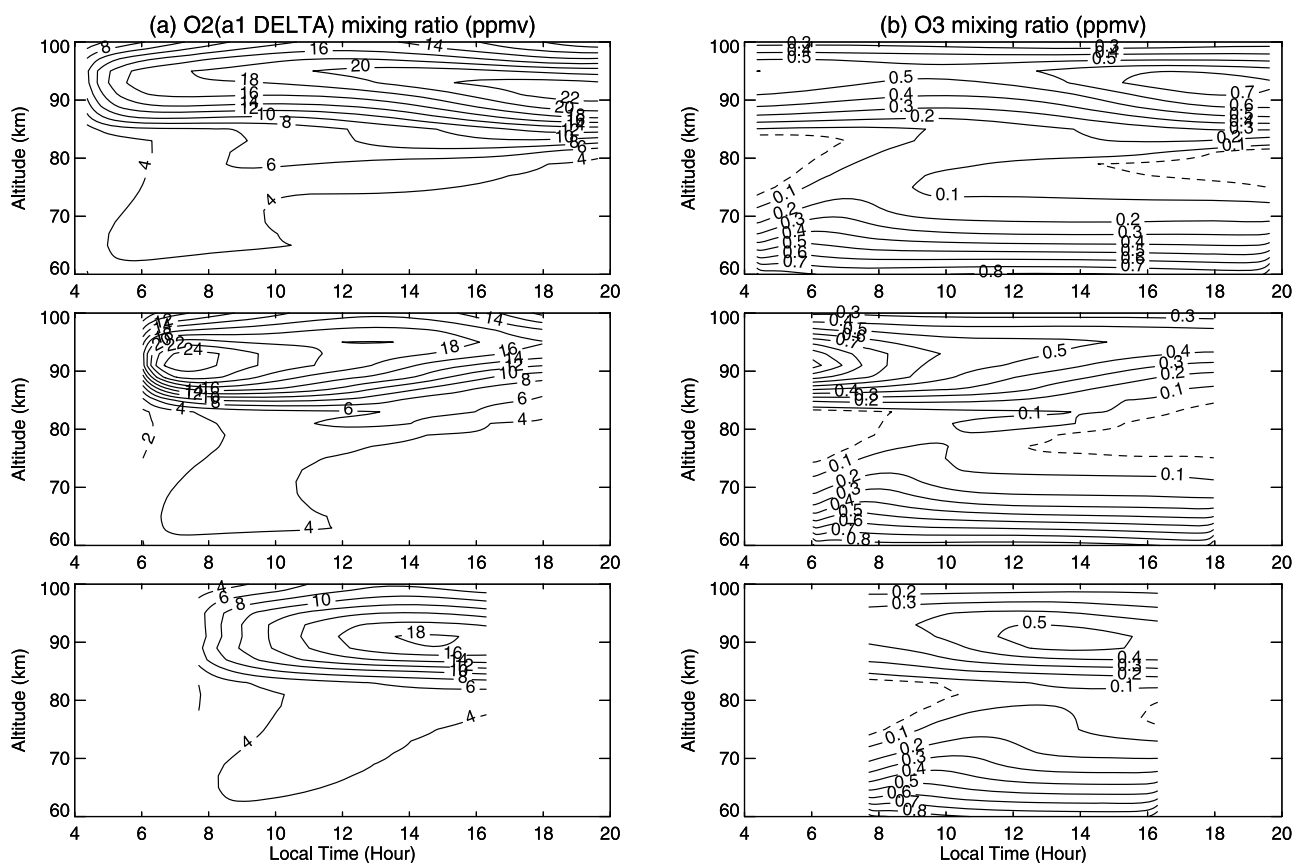


Figure 4. Same as in Figure 3 except the tidal effects have been included in the model.

Figures 1 and 2 suggest that this is a good assumption for cases with and without the effects of tides in the model except near the mesopause and twilight. Physically, it means that different models lead to different O₃ distributions that may still make only a small difference while comparing their mean values. Additional comparisons in S_{13} between different models over a larger spatial and temporal domain will be shown in the next section.

3. Model Results

[19] In Figure 3, we show the modeled daytime O₂(¹Δ) and O₃ mixing ratios on 15 June as a function of altitude and local time at three latitudes: 40°N, equator, and 40°S. The effects of tides are not included in this run, and the ozone concentrations only show small dependence on the local time. The modeled ozone near local noon is consistent with the measurements [Zhu *et al.*, 1999a]. Since it is near the Northern Hemisphere summer solstice, the daytime local time duration at 40°N is about twice that at 40°S. One distinct feature shown in Figure 3 is the mismatch of the timing of the peak O₂(¹Δ) and peak O₃ above 90 km. For example, the distinct peak in O₂(¹Δ) around 95 km occurs about 8 hours after sunrise whereas O₃ shows very small local time dependence. This is mainly caused by the photochemical interaction among various species with different timescales. Because the radiative relaxation time of O₂(¹Δ_g) is 1.24 hours and collisional relaxation becomes less important at higher altitudes, the photochemical interaction thus leads to significantly different variations in

O₂(¹Δ) and O₃ at a timescale of a few hours. We have adopted a recent value of $2.24 \times 10^{-4} \text{ s}^{-1}$ for the Einstein coefficient A_{21} [Lafferty *et al.*, 1998; Spalek *et al.*, 1999] in the model's standard runs. Our model was also tested with a smaller A_{21} of $1.47 \times 10^{-4} \text{ s}^{-1}$ used by Mlynarczyk and Nesbitt [1995] in their sensitivity study, and we found that the peak O₂(¹Δ_g) around 95 km would be increased by about 40% when the smaller value of A_{21} is used in the model, leading to a more significant difference between O₂(¹Δ) and O₃ variations. Below 80 km, however, there exists a very good correlation between the two peaks in local time: both peaks occur about 3 hours after sunrise, mainly because of the very short photochemical relaxation times in both O₂(¹Δ) and O₃. Therefore, solely on the basis of the local time variation of these two species shown in Figure 3, one may expect that a steady state model will lead to a greater fractional error in retrieved O₃ at higher altitudes. Also note that the region of both the minimum value and maximum vertical gradient of O₃ is around 80–85 km, which should lead to maximum errors in the retrieved O₃ based on the steady state photochemical model because the neglected terms in equation (13) are expected to become significant.

[20] Figure 4 shows the same plot as Figure 3 except that the tidal effects have been included in the model. Both the diurnal and semidiurnal tidal waves are included in the model [Zhu *et al.*, 2000]. Compared to Figure 3, we note that below ~80 km there is not much difference in O₂(¹Δ) and O₃ because the amplitudes and thus the effects of the

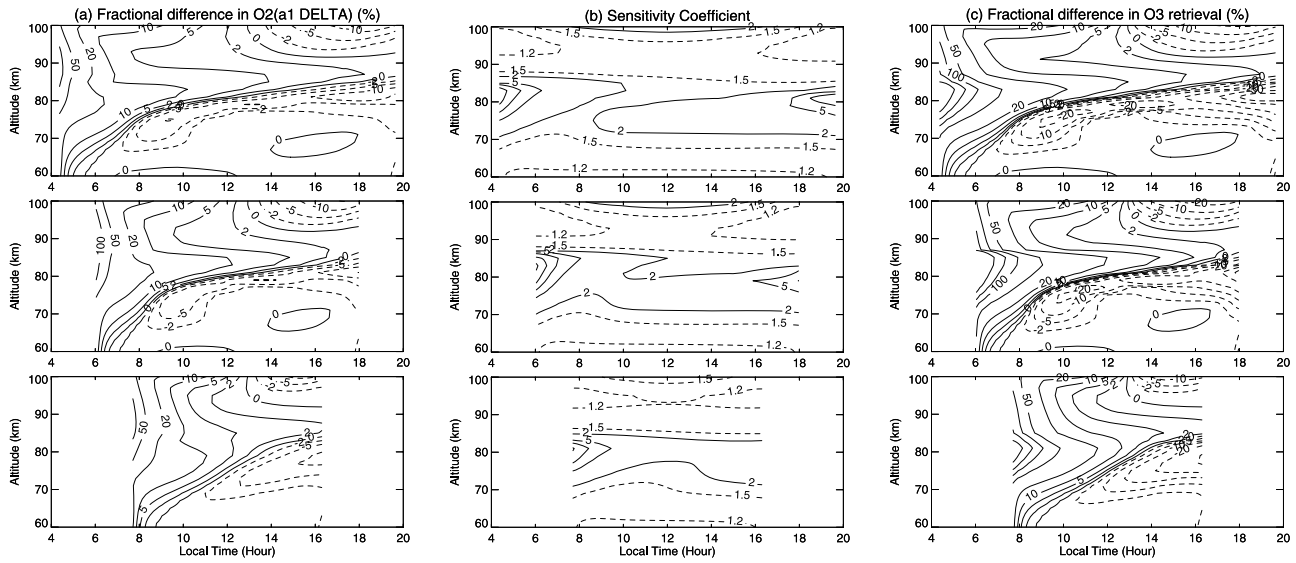


Figure 5. (a) Fractional difference of O₂(¹Δ), (b) sensitivity coefficient S₁₃, and (c) fractional difference of O₃ for the modeled daytime concentrations without tides on 15 June at three latitudes: (top) 40°N, (middle) equator, and (bottom) 40°S. The contour intervals increase with the magnitudes. In Figures 5a and 5c, the magnitude of the contours is set at 0, 2, 5, 10, 20, 50, 100, 200, 500. In Figure 5b, it is set at 1.2, 1.5, 2, 5, 10, 20, 50.

tides are small. Above ~90 km, however, both the magnitudes and peak local times change greatly because of advective transport of the species and the effect of the temperature variation on the rate coefficients [Zhu *et al.*, 2000]. It is also noted that the peak O₂(¹Δ) and peak O₃ above 90 km coincide better in local time in Figure 4 than in Figure 3. This represents the fact that tidal waves play a major role in determining the local time distributions of O₂(¹Δ) and O₃. As a result, the great difference in O₂(¹Δ) and O₃ above 90 km between Figures 3 and 4 indicates that the accurate retrieval of O₃ from O₂(¹Δ) will significantly depend on the reduction in the uncertainties in tides above 90 km.

[21] Figure 5 shows the sensitivity coefficient S₁₃ and the fractional differences of O₂(¹Δ) and O₃, $\delta[\text{O}_2(^1\Delta)]/[\text{O}_2(^1\Delta)]$, and $\delta[\text{O}_3]/[\text{O}_3]$, for the modeled daytime concentrations without tides on 15 June at three latitudes: 40°N (Figure 5, top), equator (Figure 5, middle), and 40°S (Figure 5, bottom), respectively. Here, $\delta[\text{O}_2(^1\Delta)]$ is defined as the difference between the [O₂(¹Δ)] that is calculated with the fully time-dependent model and the [O₂(¹Δ)] that is calculated from the steady state model (4)–(6) for O(¹D), O₂(¹Σ), and O₂(¹Δ), with the rest of species derived from the time-dependent model. The O₃ fractional difference $\delta[\text{O}_3]/[\text{O}_3]$ in Figure 5 is calculated according to equation (7) from $\delta[\text{O}_2(^1\Delta)]/[\text{O}_2(^1\Delta)]$ and S₁₃. Figure 5

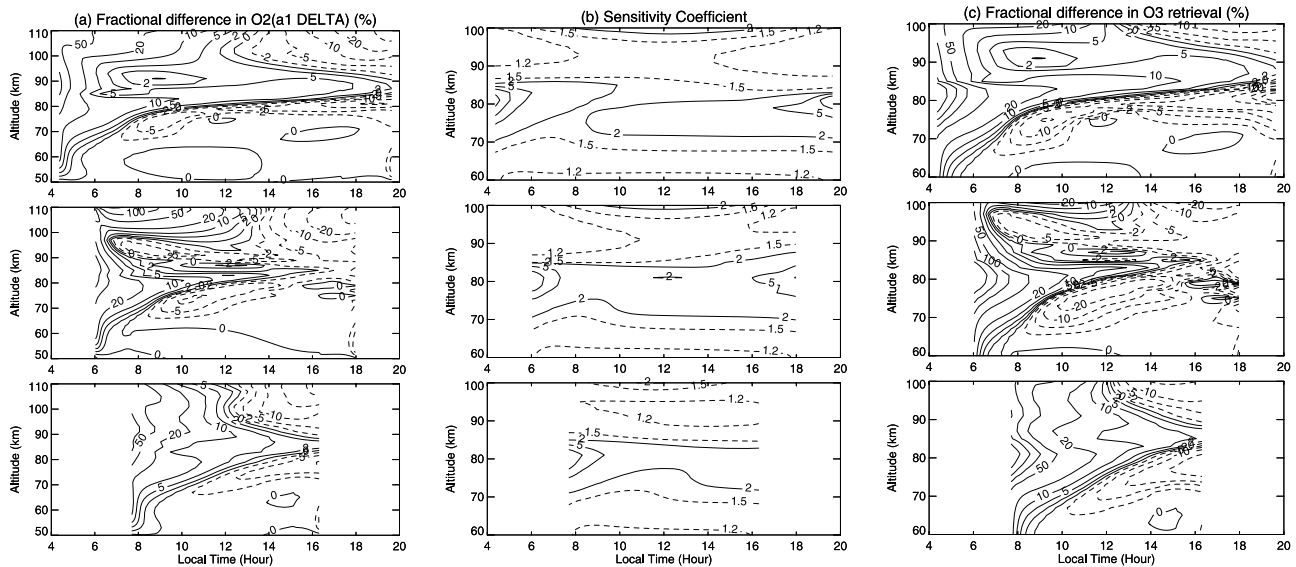


Figure 6. Same as in Figure 5 except the tidal effects have been included in the model.

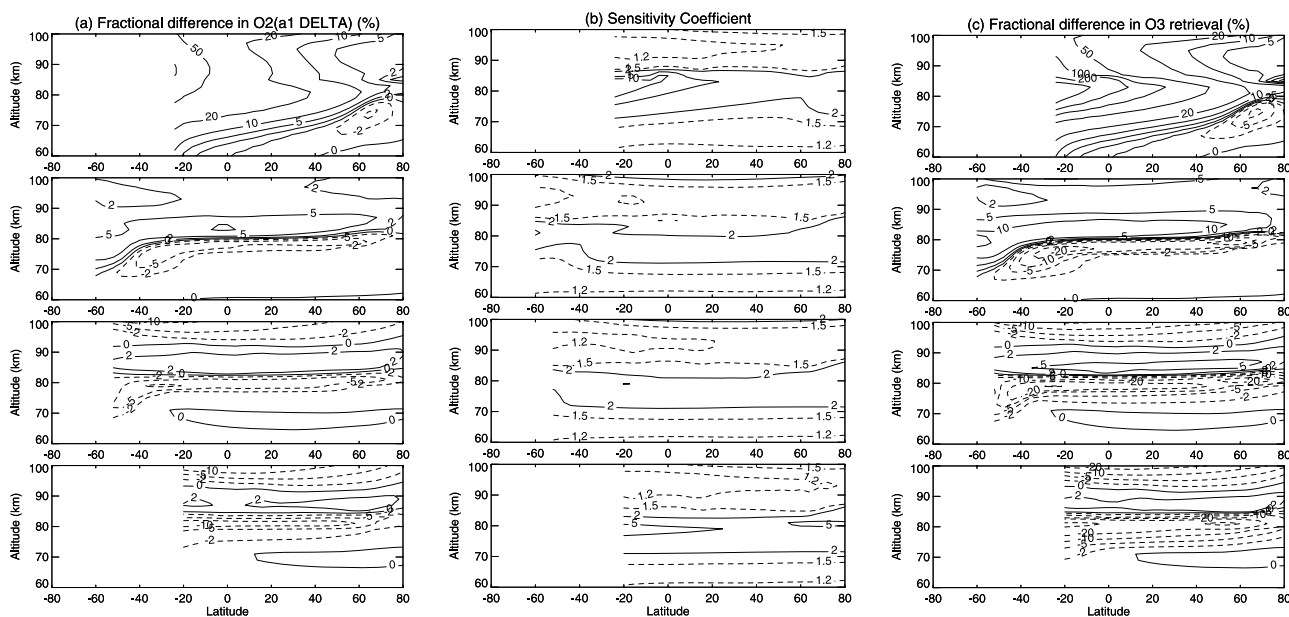


Figure 7. (a) Fractional difference of $O_2(^1\Delta)$, (b) sensitivity coefficient S_{13} , and (c) fractional difference of O_3 for the modeled daytime mixing ratios without tides on 15 June at four local times of 0700, 1200, 1500, and 1700 from top to bottom, respectively.

shows a strong local time dependence of all three fields with an overall trend of $\delta[O_2(^1\Delta)]/[O_2(^1\Delta)]$ or $\delta[O_3]/[O_3]$ decreasing with an increasing duration of the local time from sunrise. Within the first 4–6 hours after sunrise the O_3 fractional difference near the mesopause rapidly decreases from more than 100% to less than about 20%. This is mainly caused by the photochemical interaction between $O_2(a^1\Delta_g)$ and other species due to a relatively long photochemical relaxation time for $O_2(a^1\Delta_g)$. Note that in the winter hemisphere, the daytime duration is short. As a result, the local time coverage over which the retrieved O_3 has relatively small errors is reduced. Figure 6 shows the same results as in Figure 5 except that the effects of tidal waves have been included in the model runs. Comparison between Figure 5 and Figure 6 shows again that both the pattern and the magnitude of the fractional differences and sensitivity coefficient show little difference below 80 km because of the small amplitudes of the tides. Above 90 km, however, the fractional differences of $\delta[O_2(^1\Delta)]/[O_2(^1\Delta)]$ or $\delta[O_3]/[O_3]$ change significantly, whereas the sensitivity coefficient remains nearly invariant. The very limited regions that show significant changes in the sensitivity coefficient are near the mesopause and twilight, which was caused by the combination of both the small value and the large vertical gradient in $[O_3]$ mixing, as discussed before. We have already indicated that the quasi-invariance of the sensitivity coefficient with model is the theoretical basis allowing us to use equation (7) to derive $\delta[O_3]/[O_3]$ from $\delta[O_2(^1\Delta)]/[O_2(^1\Delta)]$. The results shown in Figures 5 and 6 further confirm the validity of our approach.

[22] A strong local time dependence of fractional errors in the retrieved $[O_3]$ with a significant magnitude by assuming a steady state photochemical model as shown in Figures 5 and 6 suggests that one should be cautious while using the ozone database in the upper mesosphere and lower thermo-

sphere that are retrieved from the O_2 1.27 μm airglow emissions. This has long been recognized by the community [e.g., Sica and Lowe, 1993; Gumbel et al., 1998], such as those who worked on the retrievals of O_3 by the Sounding of the Atmosphere using Broadband Emission Radiometry (SABER) instrument onboard the Thermosphere-Ionosphere-Mesosphere Energetics and Dynamics (TIMED) satellite. Specifically, the SABER team noted from the measurements that it takes 2 to 3 hours of the local time for steady state to occur near the mesopause following sunrise.

[23] Although the retrieved daytime $[O_3]$ has significant errors near the sunrise mainly due to the relatively long photochemical lifetime of $O_2(^1\Delta)$, the chemical lifetime of O_3 is only a few minutes during daytime throughout the entire mesosphere and the lower thermosphere [e.g., Brasseur and Solomon, 2005]. As a result, the modeled daytime O_3 may vary slowly with local time and remains nearly constant during the daytime in some regions of the atmosphere [e.g., Allen et al., 1984; Zhu et al., 1999a], leading to the possibility of extrapolating the retrieved ozone at later local times to that near sunrise. It is noted that the short chemical relaxation time of O_3 means that O_3 is chemically equilibrated with other species such as O or HO_x that could have either a relatively long or short relaxation time [e.g., Brasseur and Solomon, 2005]. Under these circumstances, the local time variability of those species will determine the local time variability in $[O_3]$, which can also be seen from Figures 3 and 4. Above 85 km, the chemical relaxation time of [O] or $[O_x]$ is longer than one day [e.g., Brasseur and Solomon, 2005]. As a result, daytime $[O_3]$ shows little diurnal variation as does [O] in the absence of tidal waves (Figure 3). However, when the effects of tidal waves are included, a significant local time variation of $[O_3]$ (Figure 4) occurs mainly because of the

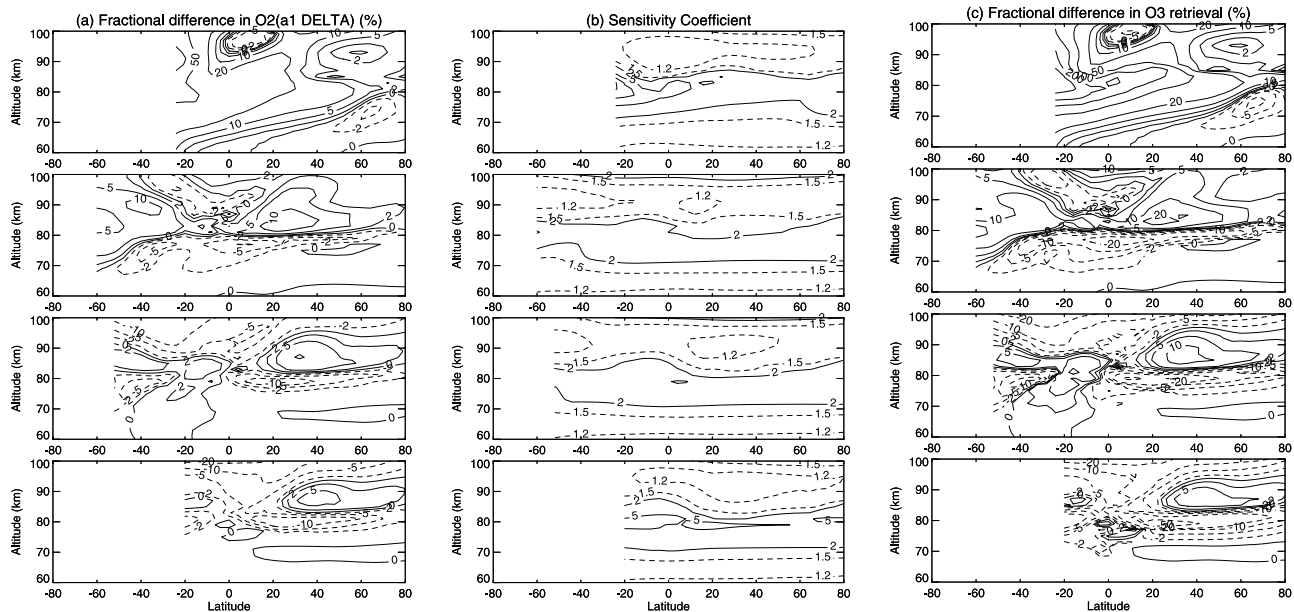


Figure 8. Same as in Figure 7 except the tidal effects have been included in the model.

vertical transport of [O] by tidal waves. Around 80 km, where [O₃] minimizes, its concentration is primarily determined by [HO_x] through the chemical loss process of H + O₃ [e.g., Allen *et al.*, 1984; Zhu *et al.*, 1999a]. Therefore modeling efforts are required to extrapolate the retrieved [O₃] from the latter local time measurements to those near sunrise.

[24] A significant set of the mesospheric daytime O₃ data was from the Solar Mesospheric Explorer (SME) satellite based on the O₂ 1.27- μ m airglow emission at a nearly fixed local time close to 1500 [Thomas *et al.*, 1984]. In Figures 7 and 8, we show the fractional differences of O₂(¹ Δ) and O₃ together with the corresponding sensitivity coefficient S_{13} at four fixed local times. We see from Figures 7 and 8 that typical errors in O₂(¹ Δ) or O₃ are about 10% or less at 1500 local time except around 80 km where there is an O₃ minimum (Figure 3). Figures 7 and 8 also show that the typical fractional errors in the afternoon are far less than those in the morning. On the basis of Figures 7 and 8, we may conclude that the typical modeling errors caused by a steady state assumption in the SME O₃ database that was retrieved from O₂ 1.27- μ m airglow emission are about 10% or less.

4. Conclusions

[25] Ozone measurements in the mesosphere and lower thermosphere can be acquired through various satellite remote sensing techniques such as direct ozone absorption in solar [e.g., Bruhl *et al.*, 1996; Chu and Veiga, 1998] or stellar [e.g., Yee *et al.*, 2002; Meijer *et al.*, 2004] occultation or ozone emission [e.g., Froidevaux *et al.*, 1996; Mlynarczyk, 1997; Riese *et al.*, 1999], or by airglow emissions of excited species such as O₂(¹ Δ) [Thomas *et al.*, 1984; Mlynarczyk, 1997] and O₂(¹ Σ) [Yee *et al.*, 1993]. There have been some discussions of advantages and disadvantages of various techniques for retrieving O₃ in the mesosphere and thermosphere [e.g., Zhu, 2004]. This paper examines errors in the retrieved O₃ based on 1.27- μ m airglow emission by O₂(¹ Δ)

in the upper mesosphere and lower thermosphere due to the dynamical-photochemical coupling associated with the relatively long chemical relaxation time. On the basis of the sensitivity coefficients that relate the directly measured volume emission rate or O₂(¹ Δ) with the inferred O₃ we are able to separate and quantify different sources of errors of the retrieval. Because of the relatively long chemical relaxation time for O₂(¹ Δ) there exists a strong local time dependence of errors in the retrieved O₃ when a steady state airglow model is used in the retrieval, such as in the case of TIMED/SABER O₃ measurements based on the O₂ 1.27- μ m airglow emissions. The fractional errors near the mesopause due to dynamical-photochemical coupling could easily exceed 50% within three hours of the local sunrise. The fractional errors generally decrease with the local time as the photochemical system slowly adjusts itself to a quasi-equilibrium state. The typical modeling errors in O₃ inferred from O₂(¹ Δ) are about 10% or less around 1500 local time when the SME satellite made its measurements of mesospheric O₃.

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